

7 The kangaroo rat is a mammal adapted to living in a warm dry habitat.

(a) The resting body temperature of a kangaroo rat is 37°C .

During exercise the body temperature is maintained between 38.5°C and 40°C.

Explain how the body temperature can be regulated to stay between 38.5°C and 40°C.

(4)

This image shows a blank sheet of white paper with ten horizontal dashed lines spaced evenly apart, resembling a template for handwriting practice or a simple ledger. The lines are thin and grey, extending across the full width of the page. There is no text or other markings on the paper.

- Suggest how kangaroo rats are adapted to survive in areas with little drinking water and dry seeds.

- | | Water balance in a kangaroo rat
(2 cm ³ / day) | Water balance in a human
(2500 cm ³ / day) |
|---------------------------------|--|--|
| Water gain
/ cm ³ | <p>food (0.2)</p> <p>metabolism (1.8)</p> | <p>food (750)</p> <p>liquid (1500)</p> <p>metabolism (250)</p> |
| Water loss
/ cm ³ | <p>faeces (0.09)</p> <p>urine (0.45)</p> <p>evaporation (1.46)</p> | <p>faeces (100)</p> <p>urine (1500)</p> <p>evaporation (900)</p> |

- (i) Calculate the difference in urine produced over seven days by the kangaroo rat and by the human.

Give your answer in standard form.

(2)

Answer

- (ii) Comment on the differences in water gain and water loss between the kangaroo rat and the human.

Use your knowledge and information in the diagram to support and explain your answer.

(4)

[illegible]

(Total for Question 7 = 13 marks)